MANYARA REGION

FORM FOUR MOCK EXMAINATION

032/2B **CHEMISTRY 2B**

ACTUAL PRACTICAL B

(*For both school and private candidates*)

TIME: 2:30 HOURS **YEAR 2021**

Instructions

1. This paper consists of **two (2)** questions. Answer **all** questions
2. Each question carries **twenty five (25)** marks
3. Qualitative analysis guide sheet authorized by NECTA and non- programmable calculators may be used.
4. Cellular phones and any unauthorized materials are **not** allowed in the examination room
5. Write your examination number on every page of your answer sheets
6. The following constant may be used.
   Na = 23, H = 1, S = 32, O = 16, Cl = 35.5, C = 12, N = 14

1. You are provided with the following solutions:
   Y: Containing 0.1M sulphuric acid
   X: Containing 0.96g NaOH in 250cm^3 of distilled water
   Procedure;
   Pipette 20cm^3 or 25cm^3 of sodium hydroxide solution into a titration flask.
   Add two dops of phenolphlthalein indicator into the titration flask.
   Titrate sulphuric acid drop-wise until an indicator changes its colour.
   Repeat the procedure to obtain three titre values
   Record your results in a tabula forms

   Questions

   (a) Calculate
   - i The volume of sulphuric acid required by 25cm^3 or 20cm^3 of sodium hydroxide
   - ii The concentration of sodium hydroxide in grams per litre
   - iii The molarity of sodium hydroxide

   (b) Calculate
   - i The number of moles of sulphuric acid required by 20cm^3 or 25 cm^3 of sodium hydroxide
   - ii The number of mole of sodium hydroxide in 25cm^3 or 20 cm^3

2. You have been provided with the following solutions
   Q : containing 0.1mol/dm^3 of Sodium thiosulphate
   Z : containing 0.5mol/dm^3 hydrochloric acid
   Distilled water
   Stop watch/clock
   Plain white paper marked X

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Procedure

i Write clear letter X on a white piece of paper

ii Place a small beaker (100cm^3) on top of the letter X such that the letter is visible when viewed from above

iii Measure 20cm^3 of **Q** and put it into the beaker in (ii) above

iv Measure 10cm^3 of **Z** and pour it into a beaker containing solution **Q** in (iii) above and simultaneously start the stop watch/clock

v Swirl the reaction mixture only once and note the time taken for the letter X to disappear completely

vi Repeat procedure (ii) to (v) by varying the volume of **Q** and distilled water as indicated in the table below.

Table: Experimental Data

Volume of solution Q (cm^3)	**Volume of distilled water (cm^3)**	**Volume of solution Z (cm^3)**	**Time (s)**	**Rate of reaction i/t (s^{-1})**
20	0	10		
15	5	10		
10	10	10		
5	15	10		

Questions

(a) Complete filling the Table

(b) Show the reaction equation between **Z** and **Q**

(c) Plot a graph of volume of **Q** against time **(s)** taken for the reaction

(d) Comment on the nature of the graph obtained.